Lifecycle Phase	Category	Code	Apply when	N	n	Example Quotes	Participants
Initial setup	Ease	Key generation easiest	Names key-pair generation as an easiest step.	46	22	“The easiest step in setting up client authentication was generating the cryptographic key pair with OpenSSL.” — P11	P5, P8, P10, P11, P13, P14, P15, P16, P17, P19, P22, P24, P27, P30, P32, P34, P35, P36, P37, P39, P40, P44
Initial setup	Ease	CA signing or download easy	Names CSR submission, signing, or certificate download as easy.	46	7	“The easiest steps in setting up client authentication were using the website to upload the certificate and downloading the signed certificate.” — P21	P1, P6, P14, P20, P21, P29, P46
Initial setup	Ease	Browser import or authentication easy	Names certificate import, browser registration, or the final test as easy.	46	12	“I didn’t even look up how to do it and just went immediately to the Firefox security settings, found the certificate section, and clicked on the import button.” — P18	P4, P12, P18, P25, P34, P35, P37, P38, P39, P40, P42, P45
Initial setup	Ease	No major change needed	Says the setup was acceptable or requests no substantive change.	46	12	“I don’t think I would change anything about the setup process, once I realized what I was missing it was fairly easy to follow.” — P13	P5, P6, P9, P11, P13, P14, P16, P20, P23, P37, P38, P46
Initial setup	Challenge	Failed or malformed CSR	Reports a failed, invalid, or malformed CSR attempt or submission.	46	35	“I got 4 errors including missing attribute, incorrect attribute, insufficient key security, and some other issue.” — P4	P1, P2, P4, P5, P7, P8, P9, P10, P11, P12, P13, P15, P16, P18, P19, P20, P22, P25, P26, P27, P29, P30, P31, P32, P33, P35, P36, P38, P39, P40, P41, P42, P43, P45, P46
Initial setup	Challenge	Unsupported key size	Reports rejection or rework caused by an insufficient or unsupported key size.	46	10	“Tried to use RSA 3072 bit key and EC 256 and 384 key, but they did not satisfy the requirements despite being examples of the required level of bit security.” — P40	P2, P9, P19, P29, P31, P32, P35, P40, P43, P46
Initial setup	Challenge	OpenSSL default method issue	Attributes failure to a default or commonly suggested OpenSSL command.	46	10	“This command would not work because it was missing part of the subject header.” — P12	P1, P12, P13, P16, P18, P19, P22, P24, P31, P36
Initial setup	Challenge	Malformed request attributes	Reports an incorrectly structured subject or request attribute.	46	12	“I created another CSR fixing the syntax error I had in one attribute and making the key longer to fix all but the missing attribute error.” — P4	P1, P2, P4, P5, P9, P12, P15, P27, P33, P38, P41, P46
Initial setup	Challenge	UID field difficulty	Reports difficulty supplying or encoding the required UID/userID value.	46	5	“Creating the proper .csr file was the most challenging part because I had never done it before, and the default settings didn't prompt for the UID, which seemed to be the biggest issue.” — P39	P10, P16, P18, P39, P45
Initial setup	Challenge	Unrecognized OID	Reports an unsupported or unrecognized object identifier.	46	2	“I had not created a new OID.” — P45	P7, P45
Initial setup	Challenge	CSR configuration or command syntax	Reports difficulty with configuration files, flags, parameters, or exact CLI syntax.	46	37	“The process of creating a configuration file to specify the UID field in the CSR was not intuitive.” — P14	P1, P2, P4, P7, P8, P10, P11, P12, P13, P14, P15, P16, P18, P19, P21, P22, P23, P24, P25, P26, P27, P28, P29, P30, P31, P34, P35, P36, P37, P38, P39, P40, P41, P42, P43, P45, P46
Initial setup	Challenge	Alternative CSR tool abandoned	Abandons a web or GUI CSR tool because required fields or compatibility are missing.	46	4	“The windows ccm doesn’t let you set the user ID to my knowledge, so I essentially abandoned this method of creating a CSR.” — P4	P4, P7, P11, P41
Initial setup	Challenge	Key–certificate association / PKCS#12	Reports difficulty bundling or associating the certificate and private key.	46	35	“Importing the certificate + private key into Chrome Chrome didn’t recognize the cert until it was in a .pfx bundle.” — P27	P2, P3, P4, P5, P6, P7, P8, P11, P12, P13, P14, P15, P16, P18, P19, P20, P21, P22, P23, P25, P26, P27, P28, P29, P31, P32, P33, P35, P36, P38, P41, P42, P44, P45, P46
Initial setup	Challenge	File format, encryption, or password	Reports PEM/CRT/P12/PFX conversion, encrypted-key, or import-password difficulty.	46	29	“When trying to add the file to my login keychain, I kept getting an error saying that the password I used to set up the file and the one I was entering did not match.” — P5	P2, P3, P4, P5, P7, P8, P11, P12, P13, P14, P15, P18, P19, P20, P21, P22, P23, P25, P27, P29, P30, P32, P36, P38, P41, P42, P44, P45, P46
Initial setup	Challenge	Browser or OS UI problem	Reports any store, import, selection, state, or compatibility problem.	46	41	“Registering the signed certificate and private key with the browser was the most difficult due to the lack of useful error feedback, which made it very difficult to diagnose issues with the setup.” — P5	P1, P2, P3, P4, P5, P6, P7, P8, P9, P10, P11, P13, P14, P15, P16, P17, P19, P20, P21, P22, P24, P25, P26, P27, P28, P29, P30, P31, P32, P33, P34, P36, P37, P38, P39, P40, P41, P42, P43, P44, P46
Initial setup	Challenge	Browser/OS registration identified as hardest	Identifies browser- or OS-level certificate import, registration, acceptance, or configuration as one of the hardest setup steps	46	18	“Figuring out what format the private keys needed to be in for the OS to accept them. Many online resources assumed you already had a .p12 file.” — P25	P1, P3, P5, P6, P7, P8, P9, P11, P15, P17, P19, P25, P27, P29, P30, P32, P36, P37
Initial setup	Challenge	Store discovery and placement	Cannot identify or verify the correct credential store or import location.	46	28	“I just imported the cert into 5-6 different cert locations on my computer.” — P16	P1, P2, P3, P4, P5, P7, P8, P10, P11, P15, P16, P17, P19, P20, P22, P24, P25, P27, P29, P30, P31, P32, P34, P36, P37, P41, P44, P46
Initial setup	Challenge	Activation, selection, or browser state	The correct credential is not offered or selected because of browser state or cache.	46	15	“I encountered several issues related to SSL cache and automatic certificate selection, which required manual intervention.” — P37	P2, P5, P7, P9, P10, P16, P17, P20, P24, P27, P31, P34, P37, P42, P43
Initial setup	Challenge	Opaque error or compatibility trap	Weak diagnostics or a version/platform mismatch blocks progress or recovery.	46	24	“The reason why this was so hard was because I was getting very minimal information as to why they were failing, and had to spend a few extra minutes troubleshooting.” — P32	P2, P4, P5, P8, P9, P10, P16, P17, P19, P24, P25, P27, P30, P32, P34, P35, P37, P38, P40, P41, P42, P43, P44, P46
Initial setup	Challenge	Installation, dependency, or version issue	Reports installation, package, build, dependency, or version difficulty.	46	8	“I attempted to use perl and NASM via the visual studio developer console to install it, but that only caused more problems than it solved.” — P42	P5, P8, P16, P19, P38, P42, P43, P46
Initial setup	Challenge	Permission or managed-device restriction	Reports administrative rights, policy, or device restrictions.	46	1	“I believe this is because of the fact that I did this project on my work computer, which potentially has some restrictions set up that was preventing me from being able to import certificates easily.” — P32	P32
Initial setup	Challenge	Sparse, conflicting, or outdated information	Reports missing, vague, conflicting, obsolete, or dead-end guidance.	46	20	“I was not able to find information on how to make it work with Google Chrome.” — P17	P1, P2, P4, P7, P8, P10, P13, P16, P17, P19, P24, P25, P27, P31, P35, P36, P41, P43, P45, P46
Initial setup	Challenge	Repeated trial and recovery	Reports repeated attempts, restarts, discarded artifacts, or trial-and-error recovery.	46	31	“After a lot of trial and error, I realized that I had to convert the certificate and key into a single .pfx file using openssl and that I had to use Chrome instead of Brave as my browser.” — P7	P1, P2, P4, P5, P7, P8, P10, P11, P13, P15, P16, P19, P20, P22, P24, P25, P26, P27, P29, P30, P31, P32, P33, P36, P38, P40, P41, P42, P43, P45, P46
Initial setup	Challenge	CLI or nontechnical-user burden	Frames command-line work or the setup as inaccessible to nontechnical users.	46	5	“Once again I don’t know that it is realistic to expect non-technical people to use OpenSSL, but it is what I am familiar with in a limited way.” — P12	P12, P18, P19, P28, P41
Initial setup	Information source	General web search	Uses a general-purpose web search without naming a narrower source type.	46	15	“I just googled a question whenever I had one.” — P3	P1, P3, P5, P7, P12, P17, P19, P24, P27, P29, P36, P39, P41, P43, P45
Initial setup	Information source	Q&A or community forum	Uses Stack Overflow, Stack Exchange, Server Fault, Reddit, or another forum.	46	25	“I went ahead and looked up how to add additional information to CSRs and came across this similar stack overflow question.” — P28	P2, P3, P4, P9, P10, P11, P16, P18, P19, P20, P22, P25, P27, P28, P29, P30, P31, P32, P33, P35, P36, P39, P42, P45, P46
Initial setup	Information source	Official OpenSSL documentation	Uses OpenSSL documentation, manual pages, command help, or the project site.	46	16	“I also used the OpenSSL documentation and help info.” — P22	P6, P9, P10, P13, P14, P15, P16, P21, P22, P32, P33, P35, P36, P42, P43, P45
Initial setup	Information source	Vendor or tutorial documentation	Uses a vendor guide, technical article, blog, standard, or tutorial site.	46	23	“I used an online tutorial to setup OpenSSL, an IBM article to generate the key pair, and various Stack Overflow questions to create the CSR file and PKCS12 file.” — P11	P1, P5, P9, P11, P13, P14, P15, P16, P25, P26, P27, P28, P29, P30, P31, P33, P34, P35, P36, P40, P42, P43, P46

Initial setup	Information source	Video tutorial	Uses YouTube or another instructional video.	46	5	“Tutorial videos to install SSL, SSL commands and importing certificates.” — P33	P1, P31, P33, P40, P44
Initial setup	Information source	AI assistant	Uses ChatGPT or another generative assistant for explanation, commands, or debugging.	46	16	“I used only ChatGPT for the whole project, it was very helpful in providing the explaining the concept and the whole process.” — P8	P6, P8, P10, P16, P18, P19, P21, P23, P24, P32, P34, P38, P39, P43, P44, P46
Initial setup	Information source	Code or configuration example	Uses a GitHub Gist, repository, or explicit code/configuration example.	46	2	“It provided a great exploitation from a theoretical point of view and provided really good and efficient code examples.” — P8	P4, P8
Initial setup	Information source	Course or institutional guidance	Uses assignment instructions, the CA site, or other provided institutional material.	46	3	“The information sources I used were the internet, the instructions given for the assignment, and some advice from the TA.” — P7	P5, P7, P39
Initial setup	Information source	Human help or prior knowledge	Uses a TA, instructor, peer, earlier coursework, or prior technical experience.	46	12	“The information sources I used were the internet, the instructions given for the assignment, and some advice from the TA.” — P7	P2, P7, P9, P10, P12, P24, P27, P34, P38, P44, P45, P46
Initial setup	Tool	OpenSSL CLI	Uses OpenSSL for any setup step.	46	46	“To complete the project, I used OpenSSL; despite never using OpenSSL, it quickly became somewhat simple.” — P11	P1, P2, P3, P4, P5, P6, P7, P8, P9, P10, P11, P12, P13, P14, P15, P16, P17, P18, P19, P20, P21, P22, P23, P24, P25, P26, P27, P28, P29, P30, P31, P32, P33, P34, P35, P36, P37, P38, P39, P40, P41, P42, P43, P44, P45, P46
Initial setup	Tool	Browser certificate interface	Uses a browser's certificate settings, manager, import flow, or testing behavior.	46	41	“Firefox browser accepted the certificate once converted to PKCS#12 format.” — P4	P2, P4, P5, P7, P8, P9, P10, P11, P12, P13, P14, P15, P16, P17, P18, P19, P20, P22, P23, P24, P25, P26, P27, P28, P29, P31, P32, P33, P34, P35, P36, P37, P38, P39, P40, P41, P42, P43, P44, P45, P46
Initial setup	Tool	OS credential manager	Uses macOS Keychain, Windows Certificate Store/MMC/IIS, or an OS import utility.	46	21	“When trying to add the file to my login keychain, I kept getting an error saying that the password I used to set up the file and the one I was entering did not match.” — P5	P1, P2, P3, P4, P5, P6, P7, P9, P11, P15, P17, P19, P21, P25, P27, P29, P30, P32, P37, P41, P46
Initial setup	Tool	Python or provided script	Uses Python cryptography or an instructor-provided script.	46	3	“Generating the cryptographic key and CSR For generating the cryptographic key I used Python’s cryptography library.” — P5	P5, P8, P45
Initial setup	Tool	Online CSR generator	Attempts a web-based CSR generator or online compiler.	46	6	“I tried to use a CSR generator on a website to create the certificates for me, but there were no options for including a user id.” — P6	P2, P6, P7, P9, P20, P29
Initial setup	Tool	Shell or development environment	Uses Terminal, Bash, WSL, PowerShell, Git Bash, Docker, or a dev container.	46	21	“To setup my key generation and CSR generation, I only had to get a Docker file set up and a dev container utilizing it.” — P16	P2, P6, P8, P10, P15, P16, P17, P19, P21, P25, P27, P29, P30, P31, P32, P36, P38, P41, P42, P43, P46
Initial setup	Tool	Other certificate utility	Uses a certificate-specific utility such as keytool, pk12util, certutil, PuTTYgen, or a GUI manager.	46	8	“I ended up trying to import using various methods with MMC and IIS before I converted to a .pfx file and imported in the browser.” — P7	P2, P4, P7, P11, P15, P27, P40, P41
Initial setup	Coping strategy	Switch tool or browser	Changes tool, browser, or implementation path after a failure.	46	20	“The windows ccm doesn’t let you set the user ID to my knowledge, so I essentially abandoned this method of creating a CSR.” — P4	P2, P4, P5, P7, P8, P9, P10, P11, P15, P19, P20, P24, P27, P34, P38, P41, P42, P43, P44, P45
Initial setup	Coping strategy	Custom configuration or command	Uses a custom config, subject flag, or revised command to supply required fields.	46	39	“I tried this way and it didn’t work, so I went the route of making a ‘csr_config’ file with the required fields and it finally worked” — P1	P1, P3, P4, P7, P10, P11, P12, P13, P14, P15, P16, P17, P18, P19, P20, P22, P23, P24, P25, P26, P27, P28, P29, P30, P31, P32, P34, P35, P36, P37, P38, P39, P40, P41, P42, P43, P44, P45, P46
Initial setup	Coping strategy	Reset browser state	Uses restart, SSL-state/cache clearing, or private/incognito mode.	46	6	“Chrome’s certificate manager – worked once the .p12 was imported, though it only prompted in Incognito.” — P31	P5, P16, P31, P37, P42, P43
Initial setup	Coping strategy	Legacy version or downgrade	Uses an older version or compatibility mode to complete import or packaging.	46	4	“Because of this, I had to use homebrew to install openssl 1.1, unlink openssl 3, and then link openssl 1.1.” — P19	P5, P19, P29, P46
Initial setup	Coping strategy	Save reusable commands or notes	Records commands or notes for later reuse.	46	1	“Since it’s a CLI tool, I could record the commands I use for future use.” — P28	P28
Initial setup	Suggested improvement	Clearer step-by-step guidance	Requests clearer, more direct, or better ordered setup instructions.	46	10	“I recognize that it was supposed to demo the real workforce but I feel like even a job would have more direct specifications.” — P10	P2, P7, P10, P21, P27, P31, P35, P41, P42, P45
Initial setup	Suggested improvement	State UID and key-size requirements	Requests explicit custom-attribute or key-size requirements.	46	10	“I wish adding an additional attribute was just an additional command flag rather than requiring a special configuration file.” — P28	P2, P7, P15, P16, P22, P28, P39, P40, P42, P45
Initial setup	Suggested improvement	Provide known-good template or commands	Requests a config template, examples, verified commands, or vetted resources.	46	7	“Providing any resources that you know work would make the whole process much smoother.” — P4	P4, P6, P12, P25, P26, P39, P42
Initial setup	Suggested improvement	Explain file formats and bundling	Requests guidance on P12/PFX/PEM/CRT formats or key–certificate bundling.	46	7	“If I had known that the certificate needed to be .p12 file format and that it is created using the private key and the crt certificate file.” — P44	P3, P8, P21, P25, P29, P36, P44
Initial setup	Suggested improvement	Improve error messages	Requests more specific CA, browser, OS, or import diagnostics.	46	4	“A page with examples and more descriptive OS error messages about what's needed for a valid cert.” — P25	P7, P25, P32, P38
Initial setup	Suggested improvement	GUI or automation	Requests a form, GUI, script, integrated workflow, or one-command setup.	46	3	“I would have some sort of website that has a bunch of input boxes, and it would just build my CSR for me.” — P18	P18, P30, P34
Initial setup	Suggested improvement	Platform or browser guidance	Requests OS-, browser-, device-, or environment-specific instructions or support.	46	7	“The process itself was fine and fairly intuitive, my main issues came from library/browser/device specific bugs.” — P5	P1, P5, P17, P24, P25, P30, P46
Initial setup	Suggested improvement	No change	Explicitly requests no setup change.	46	9	“I wouldn’t change anything about the setup process, because after I figured out the UID field everything worked pretty seamlessly.” — P16	P6, P9, P13, P14, P16, P20, P23, P37, P46
Multi-device	Approach	Synchronize existing credential	Copies or exports the existing certificate and private key to the second device.	46	24	“I copied the certificate and created a password to protect the private key.” — P41	P1, P3, P11, P12, P13, P14, P15, P16, P18, P22, P25, P26, P27, P28, P29, P30, P31, P32, P34, P35, P39, P40, P41, P44
Multi-device	Approach	Generate a new credential	Issues a distinct key pair and certificate for the second device.	46	22	“I therefore created a new CSR, I bundled my CA-signed certificate and key into pfx format which allowed me to finish the project.” — P9	P2, P4, P5, P6, P7, P8, P9, P10, P17, P19, P20, P21, P23, P24, P33, P36, P37, P38, P42, P43, P45, P46
Multi-device	Motivation	Avoid redundant setup	Chooses synchronization to avoid repeating enrollment or setup.	24	10	“I synchronized my existing certificate because I did not feel the need to redo the entire UKM part 1 on the Hydra machine.” — P1	P1, P12, P14, P16, P18, P22, P25, P27, P31, P35
Multi-device	Motivation	Transfer is less work	Chooses synchronization because file transfer appears less labor-intensive.	24	6	“I synchronized my existing certificate to the Hydra machine since it seemed to be the easiest approach, only requiring a simple file transfer.” — P11	P3, P11, P15, P26, P34, P40
Multi-device	Motivation	Synchronize after reenrollment error	Switches to synchronization after a new-credential attempt fails.	24	1	“I chose to use the existing certificate because a) it’s just easier to use what I already have and b) I was running into errors while trying to create another” — P29	P29
Multi-device	Motivation	Synchronize after GUI-forwarding difficulty	Chooses synchronization after X11 or GUI-forwarding trouble.	24	1	“If it were any more complicated, I would have likely just created a new certificate, but considering I was using Firefox with x11 forwarding I definitely wanted to minimize the amount of time and struggle spent on this project.” — P26	P26
Multi-device	Motivation	Familiar reenrollment seems simpler	Chooses a new credential because the repeated workflow is now familiar.	22	11	“I chose this approach because I already knew the steps since I did this on my personal computer.” — P21	P2, P4, P10, P17, P19, P20, P21, P24, P33, P42, P43

Multi-device	Motivation	Unaware synchronization was possible	Chooses a new credential because synchronization was not known to be available.	22	5	“I just issued a new certificate because I didn’t know how to synchronize but I knew I could issue a new certificate relatively easily and quickly.” — P43	P7, P36, P38, P43, P45
Multi-device	Motivation	Forgot export password	Chooses a new credential after forgetting a P12/PFX export password.	22	2	“Unfortunately, I can't remember the export password for the certificate.” — P9	P8, P9
Multi-device	Motivation	Restricted file-transfer access	Chooses a new credential because SCP or another transfer path is restricted.	22	1	“Otherwise, I would’ve had to scp my certificate to the machine using my personal computer, which I did not have with me.” — P6	P6
Multi-device	Motivation	Original device or files unavailable	Reenrolls because the original device, credential, or usable export is unavailable.	46	4	“Otherwise, I would’ve had to scp my certificate to the machine using my personal computer, which I did not have with me.” — P6	P6, P8, P9, P36
Multi-device	Security concern	Private-key transfer concern	Explicitly treats copying the private key as a security risk.	46	2	“However, this method requires the user to copy and send their certificate/private key to another device. The process of sending this introduces security risks as it is vulnerable to Man-In-The-Middle attacks.” — P41	P37, P41
Multi-device	Information source	Prior report, notes, or memory	Uses a prior report, saved notes, or remembered commands.	46	14	“I relied mainly on my own prior knowledge and experience with how certificates are exported and imported between browsers.” — P14	P3, P6, P7, P12, P14, P17, P18, P21, P22, P27, P41, P42, P43, P46
Multi-device	Information source	General web search	Uses a general-purpose web search.	46	5	“I used no resources besides a quick google search on the validity of copying the full certificate over.” — P31	P16, P25, P27, P31, P37
Multi-device	Information source	Q&A or community forum	Uses Stack Overflow, Stack Exchange, Server Fault, or another forum.	46	8	“I used resources from cryptostackexchange , openssl official docs , some blog posts on importing certs in Redhat linux.” — P9	P2, P4, P8, P9, P28, P30, P39, P42
Multi-device	Information source	Official or vendor documentation	Uses OpenSSL, platform, vendor, or institutional documentation.	46	11	“I used resources from cryptostackexchange , openssl official docs , some blog posts on importing certs in Redhat linux.” — P9	P1, P8, P9, P10, P15, P20, P25, P32, P35, P39, P42
Multi-device	Information source	Video tutorial	Uses an instructional video.	46	2	“I learned that x11 was what I was supposed to use, then tried to find youtube videos to figure out how to use x11, which only showed how to ssh with x11 enabled and never how to actually use it.” — P40	P33, P40
Multi-device	Information source	AI assistant	Uses a generative assistant for commands, transfer ideas, or debugging.	46	5	“I used chatGPT for ways that I could synchronize the certificate to a new machine and it give a lot of helpful tips.” — P44	P23, P24, P28, P38, P44
Multi-device	Information source	Human help	Uses a TA, peer, or classmate for clarification.	46	8	“Support and clarification from Teaching Assistants (TAs) played a pivotal role in addressing queries.” — P8	P8, P9, P10, P17, P19, P32, P38, P40
Multi-device	Method	Reuse credential or procedural knowledge	Reuses a credential, saved command, notes, report, or prior procedure.	46	35	“I used my notes from last time I completed this project.” — P21	P1, P3, P11, P12, P13, P14, P15, P16, P18, P22, P25, P26, P27, P28, P29, P30, P31, P32, P34, P35, P39, P40, P41, P44, P2, P4, P5, P6, P7, P8, P9, P10, P17, P19, P21
Multi-device	Method	Browser export and import	Exports or copies a credential package and imports it into the second browser or store.	24	24	“First I exported as a .p12 from the apple keychain access.” — P25	P1, P3, P11, P12, P13, P14, P15, P16, P18, P22, P25, P26, P27, P28, P29, P30, P31, P32, P34, P35, P39, P40, P41, P44
Multi-device	Method	SCP, SFTP, or SSH transfer	Uses secure-copy or SSH-based transfer.	46	9	“I had saved the certificate and private key files from my personal computer and sftp’d over to the tesla machine I was using.” — P35	P1, P11, P18, P22, P25, P30, P35, P39, P46
Multi-device	Method	Email transfer	Uses email to move the credential package.	46	4	“I sent an email to my own Gmail account with the certificate that I used for part one and accessed and installed it on the T esla machine.” — P44	P14, P16, P44, P46
Multi-device	Method	Cloud or Git transfer	Uses cloud storage, Drive, Git, or GitHub to move or retain files.	46	5	“I uploaded the certificate to my Google Drive” — P34	P12, P28, P34, P36, P37
Multi-device	Method	X11 or GUI forwarding	Uses or attempts GUI forwarding to access browser functions remotely.	46	7	“I figured that I would need to use X11-forwarding to use Firefox on the Tesla servers, so I tested SSH with the -X flag in the Windows 11 terminal app first.” — P28	P26, P27, P28, P29, P35, P40, P46
Multi-device	Method	Temporary HTTP transfer	Uses or attempts a temporary HTTP server for file movement.	46	1	“I tried hosting a temporary HTTP server on Hydra to transfer the .crt file but that also did not work.” — P46	P46
Multi-device	Challenge	New-credential setup problem	In the new-credential group, reports creation, import, or update difficulty.	22	11	“Creating the new certificates proved to be the hardest step.” — P45	P2, P4, P5, P10, P19, P21, P24, P33, P36, P38, P45
Multi-device	Challenge	Locate or import problem	In the new-credential group, cannot locate or import the certificate.	22	4	“It wasn’t that hard but finding the exact menu in the settings to import the certificate.” — P43	P10, P17, P36, P43
Multi-device	Challenge	Transfer or connectivity failure	In the synchronization group, reports file-transfer or connection failure.	24	5	“Securely transferring the files without exposure added complexity, and resolving handshake errors required in-depth analysis of server logs and debugging TLS configurations.” — P39	P28, P32, P34, P39, P41
Multi-device	Challenge	X11-forwarding issue	Reports a specific X11 or GUI-forwarding failure.	24	1	“The hardest step was getting the forwarded Firefox to perform well enough to interact with. All I really did though was restart the ssh connection and hope that it fixed it.” — P28	P28
Multi-device	Challenge	Temporary HTTP issue	Reports a failure with a temporary HTTP transfer method.	24	1	“I tried hosting a temporary HTTP server on Hydra to transfer the .crt file but that also did not work.” — P46	P46
Multi-device	Challenge	Pass-off connection problem	Reports difficulty connecting to the pass-off server.	24	1	“However, the link to the Pass-Off server on the “Usable Key Management (Part 2)” was not working.” — P41	P41
Multi-device	Challenge	Lab, headless, or permission constraint	Reports lab access, remote GUI, profile, permission, or system-wide installation constraints.	46	14	“The issue I had was being unable to install the ‘cryptography’ library with ‘pip’ as it wasn’t setup on the Hydra machine, nor did I have the permissions to install it.” — P45	P5, P8, P9, P13, P15, P25, P26, P28, P30, P32, P35, P40, P45, P46
Multi-device	Challenge	Browser recognition or import problem	The second browser does not recognize, select, or import the credential correctly.	46	14	“The hardest part was getting the browser to recognize the certificate as it was throwing an odd error that I could not find much support/documentation on” — P13	P1, P3, P4, P8, P9, P10, P11, P13, P14, P22, P31, P33, P34, P37
Multi-device	Suggested improvement	Improve browser experience	Primary request concerns browser import, export, selection, or profile handling.	38	13	“The only complaint I have is how the Firefox certificate import tool is not very helpful and does not provide detailed error messages.” — P22	P3, P5, P10, P11, P13, P14, P17, P18, P22, P25, P36, P39, P45
Multi-device	Suggested improvement	Clearer migration documentation	Primary request is clearer second-device or migration guidance.	38	8	“I am not sure if I would change anything other than making sure the documentation around connecting to the Hydra machines is more prevalent.” — P32	P2, P7, P16, P21, P31, P32, P42, P43
Multi-device	Suggested improvement	Centralized or automated management	Primary request is centralized, synchronized, or automated credential management.	38	4	“It would be nice to have some kind of integration with something like a password manager that can store and sync your certificates accross devices automatically.” — P12	P6, P12, P33, P35

Multi-device	Suggested improvement	Supported secure transfer	Primary request is a safer, supported transfer mechanism.	38	4	"The utilization of SCP for certificate transfer raised security concerns." — P8	P8, P9, P19, P46
Multi-device	Suggested improvement	Better platform support	Primary request is better platform, headless, or command-line support.	38	2	"Possibly make it easier to use a browser without a GUI, or to import TLS certificates into a specific Firefox profile through the command line." — P40	P26, P40
Multi-device	Suggested improvement	No change	Requests no change and expresses satisfaction with the process.	38	7	"I think there is no need to change the process as it is easy and secure right now." — P37	P15, P20, P24, P27, P34, P37, P44
Routine use	Experience	Routine-use comment analyzed	Provides an evaluable comment specifically about day-to-day authentication.	46	23	"I just clicked the link on the project page, and it took me straight to the website without any type of authentication or verification." — P3	P3, P4, P6, P7, P11, P13, P16, P17, P18, P19, P20, P21, P22, P24, P25, P27, P28, P29, P30, P31, P35, P43, P46
Routine use	Experience	Setup hard; routine use easy	Contrasts difficult setup or migration with easy routine authentication.	46	30	"Setting up TLS client authentication is much harder than using it. Setting up seems to take 45-60 minutes whereas using it is almost seamless." — P4	P3, P4, P5, P6, P11, P12, P16, P17, P18, P19, P20, P21, P22, P25, P27, P28, P29, P30, P31, P32, P34, P35, P36, P38, P39, P40, P42, P44, P45, P46
Routine use	Experience	Seamless or automatic authentication	Describes authentication as automatic, passive, one-click, or simply working.	46	25	"Firefox allowed seamless connection to the pass-off server, handling my certificate without any extra work on my end" — P22	P3, P4, P5, P7, P11, P16, P17, P18, P20, P22, P25, P27, P28, P30, P31, P32, P34, P35, P36, P39, P42, P43, P44, P45, P46
Routine use	Experience	Certificate selection prompt	Mentions selecting, confirming, or being prompted for a certificate.	46	17	"For accessing the website, all I had to do each time was select the certificate and then enter the password associated with it." — P19	P3, P7, P11, P14, P16, P17, P19, P24, P25, P28, P29, P30, P31, P32, P35, P40, P42
Routine use	Experience	Minimal active management	Reports no maintenance or interaction after setup.	46	21	"Beyond the setup process, the TLS client authentication was pretty much a passive operation I didn't have to really think about." — P45	P3, P4, P5, P7, P13, P17, P18, P20, P22, P25, P27, P30, P31, P34, P35, P36, P38, P39, P43, P44, P45
Routine use	Experience	Recurring password or prompt friction	Reports repeated keychain, certificate-password, or selection prompts.	46	4	"My Mac did require that I enter a password every time that it shook hands with the server which was annoying sometimes..." — P13	P6, P13, P19, P46
Routine use	Experience	Device or browser dependence	Reports access being tied to a particular device, browser, or profile.	46	4	"I would switch from Microsoft Edge (my default browser) to Chrome and access the server from there." — P34	P6, P21, P34, P36
Routine use	Storage	Storage practice described	Describes where or how the signed certificate and private key were stored.	46	34	"The certificate and private key were stored securely on my local computer in a dedicated folder with appropriate file permissions." — P46	P1, P2, P3, P4, P5, P6, P8, P9, P12, P13, P14, P17, P18, P19, P21, P22, P25, P26, P27, P28, P29, P30, P32, P35, P36, P37, P39, P40, P41, P42, P43, P44, P45, P46
Routine use	Storage	Integrated credential store	Uses an OS or browser credential store as the primary reported location.	34	17	"Throughout the semester, I maintained my certificate and key using the Keychain Access feature application on my MAC." — P8	P2, P3, P4, P5, P8, P9, P13, P14, P18, P19, P22, P25, P30, P32, P35, P40, P43
Routine use	Storage	macOS Keychain	Uses macOS Keychain as the primary reported location.	17	11	"Throughout the semester, I maintained my certificate and key using the Keychain Access feature application on my MAC." — P8	P2, P3, P5, P8, P9, P13, P19, P22, P25, P30, P32
Routine use	Storage	Browser certificate store	Uses a browser certificate store as the primary reported location.	17	6	"The browser also managed my private key for me, so I did not have to worry about where it was." — P14	P4, P14, P18, P35, P40, P43
Routine use	Storage	Local device folder	Keeps credential files in a local folder on the device.	34	14	"The certificate and private key were stored securely on my local computer in a dedicated folder with appropriate file permissions." — P46	P1, P6, P17, P21, P26, P27, P29, P37, P39, P41, P42, P44, P45, P46
Routine use	Storage	Private Git repository	Keeps credential files in a private Git repository.	34	3	"I had it on a private git repo but I never had it use it after the first time." — P36	P12, P28, P36
Routine use	Storage	Cloud or removable backup	Mentions cloud storage or removable media as an additional storage or backup path.	46	2	"Had it on my computer all semester, and I uploaded all my project files on my Google Drive." — P27	P27, P37
Routine use	Storage	Recognized insecure storage	Explicitly acknowledges that the chosen storage practice is insecure or poor.	46	4	"I'm aware that it's bad practice to push keys and environment variables to GitHub repositories but the repository is private and I am the only collaborator." — P28	P1, P12, P27, P28
Routine use	Recovery	Regeneration, loss, or password recovery	Reports regenerating credentials or recovering from loss or a forgotten password.	46	8	"Due to a password oversight during certificate generation, I had to regenerate my certificate and key." — P8	P8, P9, P10, P13, P22, P33, P38, P45
Understanding	mTLS vs. password	Password attack resistance	Identifies resistance to guessing, reuse, brute force, phishing, or password theft.	46	22	"By using public key cryptography with an acceptable key length, you are protected from a brute force attack and password guessing/cracking is out of the picture." — P30	P3, P4, P7, P8, P9, P12, P16, P17, P18, P20, P27, P29, P30, P32, P33, P37, P39, P41, P42, P43, P44, P46
Understanding	mTLS vs. password	Stronger cryptographic assurance	Attributes benefit to certificates, trusted issuance, mutual authentication, or strong cryptography.	46	16	"TLS client authentication provides stronger authentication than password-based authentication because it requires the client to present a digital certificate that is signed by a trusted certificate authority (CA)." — P9	P3, P8, P9, P17, P19, P21, P27, P29, P30, P32, P33, P35, P37, P39, P43, P46
Understanding	mTLS vs. password	Seamless no-password use	Identifies reduced typing or remembering and automatic routine authentication.	46	13	"The benefit of using TLS authentication compared to password-based authentication is that you do not have to enter your password every time you enter a secure site" — P1	P1, P2, P18, P22, P27, P28, P34, P35, P36, P38, P40, P44, P46
Understanding	mTLS vs. password	Key or device compromise	Recognizes impersonation risk if the private key, credential file, or device is compromised.	46	15	"Someone who has access to your device either physically or remotely could copy the certificate/key file and impersonate you." — P45	P2, P5, P6, P13, P15, P17, P18, P25, P26, P33, P34, P39, P41, P42, P45
Understanding	mTLS vs. password	Setup and lifecycle burden	Identifies setup, storage, issuance, revocation, or management complexity.	46	26	"The drawback is that certificate management is horrible." — P16	P1, P5, P7, P10, P11, P14, P16, P17, P18, P20, P21, P22, P23, P27, P29, P31, P32, P35, P36, P37, P38, P39, P41, P42, P43, P46
Understanding	mTLS vs. password	Portability or recovery burden	Identifies device dependence, cross-device use, loss, or recovery difficulty.	46	6	"Also, it only works on the computer you set it up on, and also only the browser you set it up on." — P38	P18, P20, P22, P27, P38, P41
Understanding	mTLS vs. password	No downside misconception	Claims mTLS has only benefits or no material security downside.	46	3	"TLS seems to only provide security benefits as it removes all of the potential security vulnerabilities related to passwords." — P7	P7, P24, P31
Understanding	mTLS vs. password	No added security misconception	Claims mTLS offers no security benefit beyond passwords.	46	1	"I think that it is nice that it puts less burden on the user to remember passwords but it does not feel like it gives any more security benefits." — P36	P36
Understanding	Sync vs. new certificate	Synchronization convenience	Identifies lower effort, continuity, or simpler management from synchronization.	46	25	"Synchronizing an existing certificate to a new machine is a relatively easy process." — P9	P3, P5, P7, P8, P9, P10, P11, P13, P15, P16, P18, P21, P27, P28, P29, P30, P31, P32, P35, P37, P39, P41, P42, P43, P46
Understanding	Sync vs. new certificate	Transfer exposure	Identifies interception or exposure while copying the credential or private key.	46	18	"Synchronizing an existing certificate provides continuity and ease of use, but it increases the risk of exposure during transfer." — P39	P1, P5, P6, P11, P12, P13, P17, P19, P21, P27, P33, P35, P37, P39, P40, P41, P44, P46
Understanding	Sync vs. new certificate	Shared compromise or revocation scope	Recognizes that one shared credential expands compromise or replacement across devices.	46	20	"The drawback is if one certificate is compromised, then both could be." — P3	P2, P3, P5, P8, P9, P10, P16, P20, P22, P25, P26, P27, P28, P29, P30, P31, P32, P34, P38, P43
Understanding	Sync vs. new certificate	Per-device isolation	Recognizes independent compromise, replacement, or revocation with distinct credentials.	46	26	"Issuing a new certificate eliminates this risk by isolating a certificate per machine." — P3	P3, P4, P5, P7, P8, P9, P10, P12, P16, P17, P18, P22, P25, P26, P27, P28, P29, P30, P33, P34, P35, P37, P39, P41, P42, P46

Understanding	Sync vs. new certificate	Multiple-credential overhead	Identifies increased issuance, tracking, or management work for separate credentials.	46	25	“A new cert on every machine creates operational issues. Both the number of keys increases and the headache to manage them increases.” — P27	P4, P5, P7, P8, P9, P16, P17, P18, P21, P25, P26, P27, P28, P30, P32, P33, P35, P37, P38, P39, P41, P43, P44, P45, P46
Understanding	Sync vs. new certificate	Incorrect or absent trade-off	Treats shared and separate credentials as equivalent or gives reversed reasoning.	46	1	“Its probably more secure to use the same private key because it is one less password that can be used to access the site” — P23	P23
Understanding	Lost certificate; key intact	Reissue with retained key	Proposes a new CSR or certificate using the retained private key.	46	36	“You could create another certificate signing request using the same key.” — P6	P2, P3, P4, P5, P6, P7, P8, P9, P11, P12, P13, P14, P15, P16, P17, P18, P19, P22, P23, P24, P25, P26, P27, P30, P31, P32, P33, P34, P35, P37, P39, P41, P43, P44, P45, P46
Understanding	Lost certificate; key intact	Unnecessary key replacement	Replaces the intact key without a stated compromise reason.	46	5	“I am not too sure, but I would probably just establish a new TLS certificate and private key pair.” — P20	P20, P21, P22, P25, P38
Understanding	Lost certificate; key intact	Revoke lost public certificate	Proposes revoking or invalidating the lost certificate as if loss alone enabled use.	46	11	“If you were to lose the certificate but not the private key, then you would need to first contact the CA and tell them to revoke that certificate.” — P18	P5, P18, P19, P20, P21, P23, P26, P27, P31, P35, P39
Understanding	Lost certificate; key intact	Certificate loss treated as compromise	Treats loss of the public certificate as actual or possible credential compromise, despite the scenario stating that the private key remains secure.	46	10	“If they do not, then an attacker could use that certificate to pretend they are you.” — P18	P8, P9, P14, P18, P19, P21, P26, P31, P34, P39
Understanding	Lost certificate; key intact	Reissuance identity verification	Raises identity proofing or unauthorized reissuance as the main concern.	46	10	“There could be a security concern if someone can try to impersonate me during the issuance process.” — P17	P9, P12, P17, P24, P27, P33, P37, P41, P42, P43
Understanding	Lost certificate; key intact	Unknown or no workable recovery	Provides no actionable recovery path or proposes reconstruction without a basis.	46	5	“I’m not sure, but I believe I saw that there's a way to either reverse engineer your certificate or to recreate it?” — P42	P1, P10, P29, P36, P42
Understanding	Lost private key	Generate new key and certificate	Proposes a new key pair, CSR, and replacement certificate.	46	38	“I would need to create a new key pair to generate a new CSR. Then, use the csr to create a new certificate.” — P34	P3, P4, P5, P6, P7, P8, P9, P11, P12, P13, P14, P15, P16, P17, P18, P19, P20, P21, P22, P23, P25, P27, P28, P30, P31, P32, P33, P34, P35, P36, P37, P38, P39, P41, P43, P44, P45, P46
Understanding	Lost private key	Revoke old certificate	Proposes revoking or invalidating the certificate bound to the lost key.	46	23	“The certificate must be immediately revoked, the key-pair must be destroyed, and the entire creation process must be redone.” — P27	P3, P5, P6, P8, P9, P11, P17, P18, P19, P21, P23, P26, P27, P28, P31, P32, P33, P35, P39, P41, P43, P45, P46
Understanding	Lost private key	Loss treated as active compromise	Assumes a merely lost key is already controlled by an attacker.	46	6	“Any attacker with access to the lost private key could impersonate you until the associated certificate is revoked.” — P9	P3, P8, P9, P17, P18, P19
Understanding	Lost private key	Private key recoverability misconception	Claims the key can be recovered from a certificate, CSR, password, or public artifact.	46	4	“I'm almost certain I saw something about reverse engineering a csr and private key from a crt file, so assuming you still have that I assume it's possible?” — P42	P2, P26, P40, P42
Understanding	Lost private key	Identity verification concern	Raises identity proofing during replacement as a concern.	46	6	“Similar to the above solution, I think some two-factor authentication would be necessary.” — P24	P12, P18, P24, P25, P31, P41
Understanding	Lost private key	Unknown or no response	States uncertainty without a workable recovery action.	46	3	“Don't Know” — P29	P1, P10, P29
Understanding	Stolen private key	Revoke, notify, or block	Calls for prompt revocation, blocking, or notification of the CA or administrator.	46	35	“Similar to losing the private key, inform the CA to revoke the certificate, generate a new key pair, and request a new certificate.” — P9	P2, P3, P5, P7, P8, P9, P10, P11, P12, P13, P15, P16, P17, P18, P19, P20, P21, P22, P23, P25, P26, P28, P30, P32, P33, P34, P35, P37, P38, P39, P40, P41, P42, P43, P46
Understanding	Stolen private key	Generate replacement credential	Calls for a new key pair and certificate after theft.	46	34	“The compromised certificate would have to be revoked and a new certificate and new private key would have to be issued.” — P3	P2, P3, P4, P5, P6, P8, P9, P10, P11, P13, P14, P15, P16, P18, P20, P21, P22, P23, P25, P26, P27, P28, P30, P32, P33, P34, P35, P36, P39, P41, P43, P44, P45, P46
Understanding	Stolen private key	Active impersonation risk	Recognizes that the thief can authenticate or impersonate until containment.	46	18	“If the private key is stolen, immediate revocation of the associated certificate is necessary to prevent unauthorized access.” — P39	P7, P8, P9, P12, P17, P18, P19, P24, P25, P26, P27, P31, P33, P35, P36, P39, P41, P43
Understanding	Stolen private key	Identity verification or recovery abuse	Raises fraudulent recovery, revocation, or identity-proofing risk.	46	8	“someone could pretend to be you and blacklist your legitimate one.” — P2	P2, P10, P12, P18, P22, P24, P31, P42
Understanding	Stolen private key	Treats theft like passive loss	Gives the same recovery as accidental loss without accounting for attacker control.	46	5	“You would need to take the same approach as thought exercise 4.” — P45	P4, P6, P11, P44, P45
Understanding	Detection and revocation	Behavior, IP, device, or log monitoring	Uses external behavioral, location, device, or audit signals to flag misuse.	46	34	“Identification Measures: Monitor for unusual access patterns, such as login attempts from different geographical locations.” — P8	P1, P2, P3, P4, P5, P7, P8, P9, P11, P12, P14, P16, P17, P18, P19, P20, P22, P24, P25, P26, P30, P31, P32, P33, P34, P35, P37, P38, P39, P40, P41, P43, P45, P46
Understanding	Detection and revocation	User or administrator reporting	Relies on reports or observed consequences to initiate investigation.	46	7	“Another way this could be found out is when students start complaining about their grades to the professor” — P18	P18, P22, P29, P32, P34, P35, P41
Understanding	Detection and revocation	Revocation, blocklist, CRL, or OCSP	Uses certificate revocation or server-side blocking to stop further use.	46	33	“The CA authority, if it can identify a stolen certificate, could then invalidate the certificate to ensure it cannot be used.” — P19	P1, P2, P3, P4, P5, P8, P11, P12, P13, P16, P18, P19, P20, P21, P22, P25, P26, P27, P28, P29, P30, P31, P32, P33, P34, P35, P37, P38, P39, P41, P43, P45, P46
Understanding	Detection and revocation	Additional MFA or device binding	Proposes MFA, device binding, hardware protection, or another control beyond the certificate.	46	17	“Additionally, the CA could employ multi-factor authentication or other advanced security measures for sensitive databases to reduce the risk of unauthorized access, even if a certificate is stolen.” — P37	P3, P5, P7, P10, P12, P14, P19, P20, P23, P24, P25, P26, P30, P36, P37, P42, P44
Understanding	Detection and revocation	External-signal limitation	Recognizes that stolen use is not intrinsically distinguishable without reports or context.	46	4	“I believe the Certificate Authority would have no way of differentiating a valid certificate from a compromised certificate.” — P22	P18, P22, P28, P29
Understanding	Detection and revocation	Cryptographic detection misconception	Assumes ordinary signatures, passwords, or challenges reveal that a valid credential was stolen.	46	3	“I think using a digital signature could prevent this attack.” — P6	P6, P21, P27